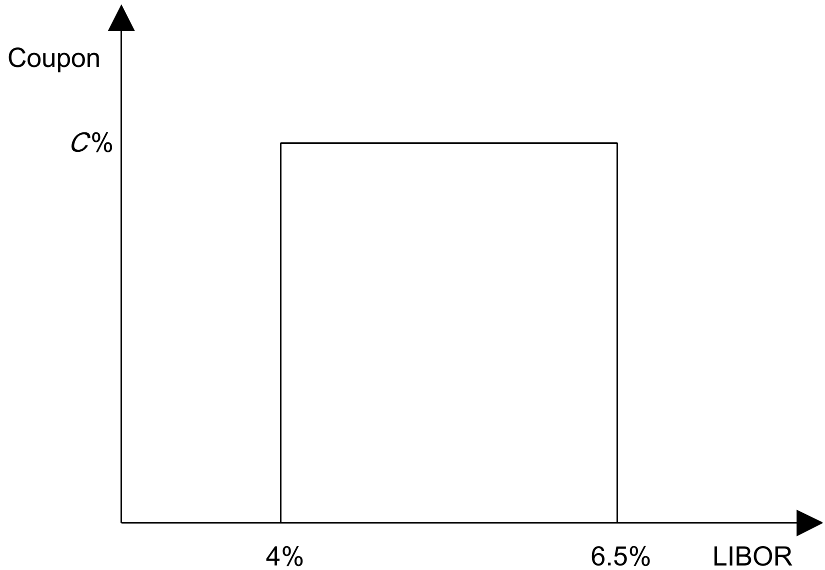

Figure 7.10: Coupon Paid Each Year in the Binary Accrual Note

We need to iterate for that value of C such that the value of the binary accrual coupons is equal to the value of the LIBOR flows that the issuer pays. This is shown in Figure 7.11, with an assumed constant base vol of 15%. As shown, we solve for a binary accrual coupon C equal to 6.786%, a pickup of 131.6 bps over the 4-year swap rate.

Price Sensitivity of the Binary Accrual Note

Figure 7.12 shows the price sensitivity of the binary accrual note to instantaneous parallel shifts in the swap curve.[28] This bond, which is purchased for par, will only trade above par for a relatively narrow band of shifts in the curve.[29] In fact, if the market sells off at all, this bond starts trading below par. However, the bond can sustain a substantial rally and will still trade above par. What is the reason for this? This bond is essentially a fixed rate bond but is one in which the investor has sold options that will wipe out the

[28]We have made the assumption that the first coupon of the bond remains set at $C = 6.786\%$ as we shift the swap curve.

[29]It makes intuitive sense that this bond will only trade above par provided that rates don't move that much. The owner of this bond has sold options (i.e., a binary put and a binary call) in order to enhance his yield. The bondholder will benefit provided that the underlying does not move much, as is often the case when one sells options.

TIme	zc	Period	Forward	σ	$d_2[X_1]$	$d_2[X_2]$
1	5.000%	0x1	5.000%	n/a	n/a	n/a
2	5.113%	1x2	5.226%	15%	1.7070	-1.5297
3	5.312%	2x3	5.712%	15%	1.5736	-0.7151
4	5.494%	3x4	6.041%	15%	1.4572	-0.4115

TIme	$N(d_2[X_1]) - N(d_2[X_2])$	Binary coupon	Float	PV Float
1	n/a	(6,462,740)	5,000,000	4,761,905
2	0.8930	(5,484,891)	5,225,748	4,729,739
3	0.7049	(4,095,632)	5,712,202	4,890,656
4	0.5871	(3,216,829)	6,041,364	4,877,792
		(19,260,092)		19,260,092
		NPV	0	

X_1	4.000%
X_2	6.500%
C	6.786%
Notional	100,000,000

Figure 7.11: Solving for the Fixed Coupon in the Binary Accrual Note

coupons if the market moves too much in either direction. If the market sells off, this impacts the value of the bond in two ways. First, as with most any fixed rate bond, a sell-off will diminish the value of the bond.[30] Second, given where the forwards are, as the market sells off, this will increase the chance that the upper boundary is broken, which will send the coupons to zero. Both of these effects serve to decrease the value of the bond if the market sells off.

Let's examine now the effect of a rally on the value of this binary accrual note. A slight rally in rates has two effects on this bond. First, the value of the fixed rate coupons will tend to go up, as is the case with most any fixed rate

[30] One notable exception is an IO bond, a mortgage security that often increases in value as the market sells off.

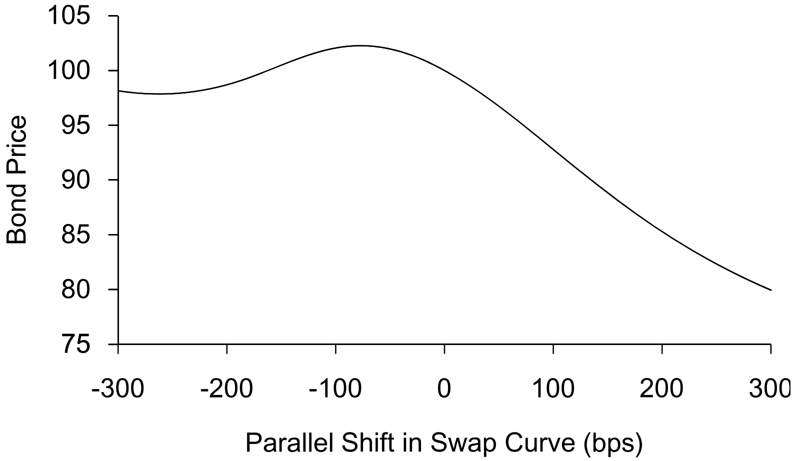

Figure 7.12: Price Sensitivity of the Binary Accrual Note for Instantaneous Shifts in the Swap Curve

bond. Second, a slight market rally serves to pull the forwards back toward the middle of the range of 4% to 6.50%, decreasing the probability that the coupons will be knocked out. Both of these effects serve to increase the value of the bond for a slight market rally.

Next, suppose that the market rallies a lot. While the initial rally raises the value of a fixed rate bond, a substantial decline in rates increases the probability that LIBOR will set below the lower boundary of 4%, resulting in a decline in the value of the bond. If rates fall enough, this latter effect outweighs the former, and the bond will trade below par and decline in value as the market rallies further. For an extreme rally (corresponding to the left-most region of Figure 7.12), we see that the price of the bond once again starts to increase as rates fall. When rates fall enough, the probability that LIBOR sets below the lower boundary becomes quite high, and the bond begins to behave like a zero coupon bond. Any further rally increases the value of the bond, just as a zero coupon bond increases in value as the market rallies.

We can use the information implicit in Figure 7.12 and Equation 7.5 to compute the effective duration of the bond. Starting at the extreme left of Figure 7.13, we see that when rates are extremely low, the binary accrual note has positive duration. In this region, the market has rallied so far that there is virtually no probability that the bondholder will ever receive the high coupon C, and the bond trades like a zero coupon bond, which has positive

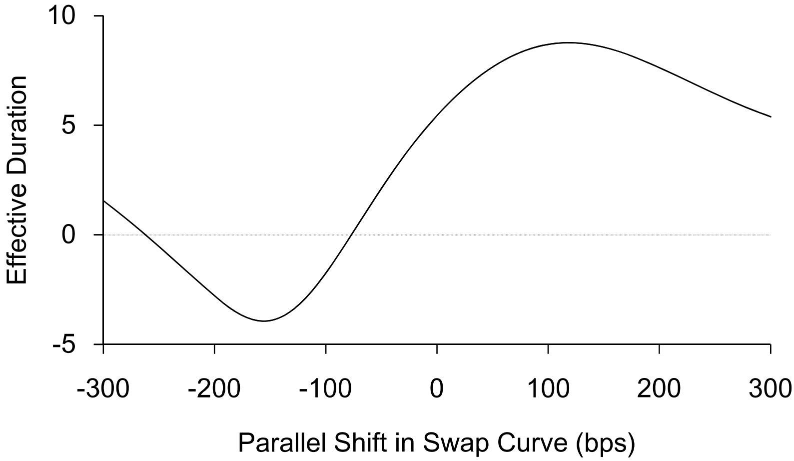

Figure 7.13: Effective Duration of the Binary Accrual Note

duration. As rates begin to rise, the probability that some of the LIBORs may set in the range rises and boosts the value of the bond. Thus, in this region, as the market sells off, the bond actually increases in value, which is to say the bond has negative duration. However, as we continue to move to the right, we see that the duration once again flips sign and becomes positive. Beyond a certain rate level the probabilities that the LIBORs finish in the range decline, and the bond decreases in value. In addition, in this range we also suffer from the effect that an increase in rates diminishes the value of the fixed coupons. Any further move to the right both decreases the probability that the LIBORs finish in the range and the value of any fixed coupons received, which decreases the value of the bond. Thus, once again the duration of the bond turns positive, and it remains positive for any higher rate levels. □

7.9 Regulatory Response

Even prior to the bankruptcy of Orange County in December 1994, regulators became concerned about the negative impact that structured notes were having on a variety of financial institutions. In the spring and summer of 1994, a number of money market funds had substantial losses due to their investment in structured notes. There was concern that some of these funds would "break the buck," meaning that the funds' shares might fall below a $1 net asset value.

The SEC wrote a letter on June 30 of that year, warning money market funds about structured notes in general and specifically about investment in five types of notes including inverse floaters, dual-index floaters, and range bonds. In July of that year, the Office of the Comptroller of the Currency (OCC) wrote an Advisory Letter known as AL 94-2. This letter was intended for regional and community banks and warned against a variety of structured notes that were believed to be "inappropriate investments for most national banks." The Federal Reserve followed with its own Supervision and Regulation (SR) Letter in August, SR 94-45, in which it indicated that certain structured notes could be acceptable investments for banks but warned banks to understand the risks inherent in structured notes. See Peng and Dattatreya (1995) for more information.

The years that followed saw lighter volumes of structured note issuance. Many investors shied away from the product in no small part due to the regulatory environment. Although some investors eventually returned, over the years new investors came into the market as well. Over the past 10 years, increased buying of dollar-denominated structured notes has been seen from both overseas investors as well as retail investors.

7.10 Non-Inversion Notes

We will end this chapter by looking at a structured note that has been very popular in recent years. *Non-inversion notes*, also called *CMS spread range accrual notes*, are structured notes in which the investor is paid an above market coupon, provided that the difference between two specified CMS rates remains greater than some prespecified amount. Consider, for example, the terms of the bond in Table 7.12.

This bond has a maturity of 15 years. It pays the investor a coupon of 8.50% in the first year. In subsequent years, the bond accrues a coupon of 8.50%, provided that the difference between 30-year CMS and 10-year CMS is not inverted. On days where 30-year CMS $-$ 10-year CMS < 0, the bond accrues a coupon of 0%. This bond is callable by the issuer starting in one year and semiannually thereafter.

Starting in around 2005, non-inversion notes became a very popular choice with investors who bought structured notes. Popular structures have had coupons that depended on the difference between 30-year CMS and 10-year CMS (as does the one in Table 7.12), as well as on the difference between 10-year CMS and 2-year CMS. Investors were attracted to the high coupons that these bonds paid. Moreover, investors were comfortable with the risks that they were taking by buying non-inversion notes. Despite the tremendous flattening and eventual inversion seen in the Treasury curve in

Issuer:	U.S. Corporate	
Principal:	$100mm	
Maturity:	15 years	
Coupon:	Year 1:	8.50%
	Year 2–15:	8.50% each day 30-year CMS − 10-year CMS ≥ 0
		0.00% otherwise
Payments:	Semiannually	
Day Count:	30/360	
Resets:	Daily for 30-year CMS and 10-year CMS for purposes of calculating the number of days the coupon is accrued	
Issue Price:	100%	
Call Feature:	This bond is callable in one year and semiannually thereafter	

Table 7.12: Terms of a Non-Inversion Note

the mid-2000s, the swap curve had never inverted since detailed records of swap rates had been kept in the early 1990s. As the Treasury curve inverted, swap spreads adjusted (with long-end spreads widening relative to front-ends spreads) so that the swap curve did not invert.

The swap curve did experience significant flattening during this time, and this made the pricing of new non-inversion note even more attractive. With a flatter swap curve, the difference between CMS rates on a forward basis declined as well. Thus, pricing models attributed a greater chance that the curve would invert in the future, which meant that the coupons that could be offered in such bonds increased.

The swap curve did eventually invert in 2006. For example, the 2s10s swap curve (the difference between the 10-year swap rate and the 2-year swap rate) inverted for the first time in February of that year and did again in November. But, as shown in Figure 7.14, the inversion was short-lived.[31]

7.10.1 The Pricing of Non-Inversion Notes

As with most all structured note issuance, the issuer enters into a swap when issuing a non-inversion note so as to create synthetic floating rate funding. Referring to Figure 7.2, the issuer and a swap dealer enter into a swap transaction in which the dealer pays the coupons on the bond to the

[31] As we will discuss in Chapter 9, other parts of the swap curve inverted as well. For example, 10s30s also inverted briefly in February 2006 and subsequently experienced more substantial inversion during the financial crisis.

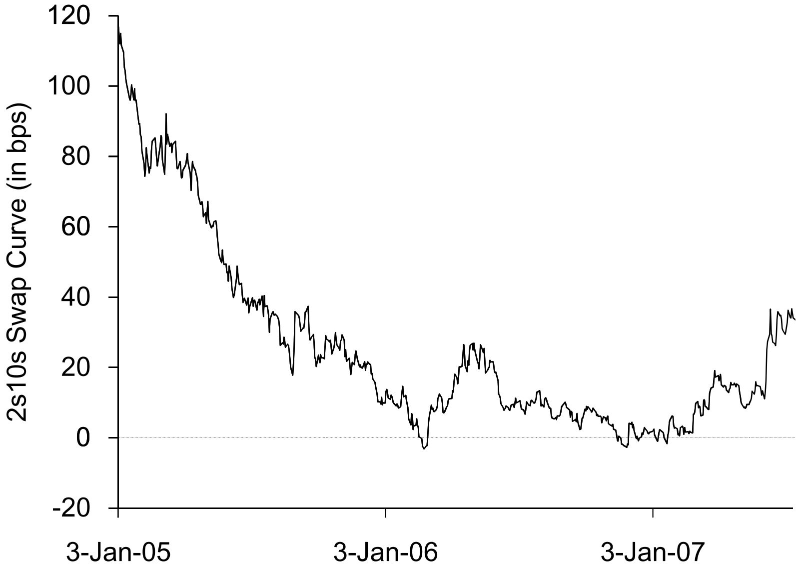

Figure 7.14: The 2s10s Swap Curve in 2005–2007

issuer, and the issuer pays 3-month LIBOR plus or minus a spread. Using the terms of the non-inversion note in Table 7.12 as an example, the two counterparties enter into a 15-year swap. The dealer pays a coupon of 8.50% in the first year and in the remaining years accrues 8.50% on the days 30-year CMS $-$ 10-year CMS ≥ 0 and 0% otherwise. The issuer pays 3-month LIBOR–based coupons. The call feature of the swap gives the swap dealer the right to cancel the swap in one year and semiannually thereafter. A representation of the underlying swap transaction to a non-inversion note can be found in Figure 7.15. Although a full pricing methodology of this swap is beyond the scope of this discussion, we can envision setting up a pricing model in which we iterate for the fixed coupon (which happens to be 8.5% in this bond) such that the underlying swap is an NPV $= 0$ transaction.

The Creation of Curve Caps

For the moment, let's ignore the cancelation option in the underlying swap depicted in Figure 7.15. For the back 14 years of the swap, the swap dealer is paying a coupon that has a daily accrual equal to

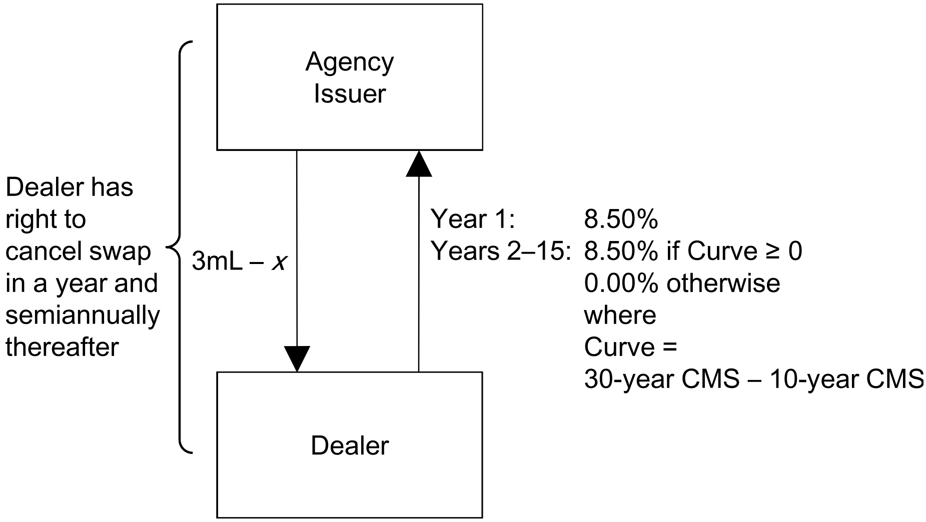

Figure 7.15: Underlying Swap Transaction Accompanying the Issuance of a Non-Inversion Note

$$\text{Coupon Payoff} = \begin{cases} 8.50\% & \text{if 30-year CMS} - \text{10-year CMS} \geq 0 \\ 0.00\% & \text{otherwise.} \end{cases} \tag{7.10}$$

Equivalently, we can regard the swap dealer paying a coupon equal to 8.50% less the payoff of a particular binary option. To this end, consider a binary option that resets daily and has a payoff as follows:

$$\text{Binary Payoff} = \begin{cases} 8.50\% & \text{if 30-year CMS} - \text{10-year CMS} < 0 \\ 0.00\% & \text{otherwise.} \end{cases} \tag{7.11}$$

Then the payoff in Equation 7.10 is equivalent to receiving 8.50% less the payoff of the binary option in Equation 7.11. To see this, consider the payoff comparison in Table 7.13.

The fact that the dealer is paying 8.50% less the payoff of this binary coupon means that the dealer is actually getting long these binary options (the ones whose payoff is given in Equation 7.11) through the issuance of non-inversion notes. These options are often referred to as *binary curve options*. Their valuation is not quite as straightforward as some

	SN Coupon	8.50% Coupon Less Binary Payout		
		8.50% Coupon	Less Binary	Difference
If 30-year CMS − 10-year CMS ≥ 0	8.50%	8.50%	0.00%	8.50%
If 30-year CMS − 10-year CMS < 0	0.00%	8.50%	−8.50%	0.00%

Table 7.13: Comparison of Structured Note Coupon and 8.50% Coupon Less Binary Option Payout

other binary options.[32] These binaries are a function of the random variable S = 30-year CMS − 10-year CMS. It would be a poor choice to assume that S is lognormally distributed, since S can assume significant negative values. In addition, the underlying random variable S is the difference between two random variables, and thus the variance of S depends on their correlation. This takes us outside the realm of anything we have looked at previously. We will discuss further the valuation of such options in Chapter 9.

The tremendous surge in the issuance of non-inversion notes over the years had meant that dealers ended up owning many of these binary curve options. There was no natural hedge for these options, and there was limited demand for these binary options from the buy-side. As dealers became long massive amounts of these options as the issuance of these notes continued, the bid for these options declined. Despite the initial lack of demand for binary options on the curve, the growth of the market of caps and floors on the difference between two CMS rates increased over time. This market eventually saw tremendous demand from investors such as hedge funds, which served as a way for dealers to lay off some of their risk. We will return to this topic in Chapter 9.

The Call Feature

The non-inversion note in Table 7.12 contains a call feature that allows the issuer to call the bond starting in one year and semiannually thereafter. This call feature is fairly common in most non-inversion notes. Of course, the issuer would normally pass on the call feature to the dealer in the underlying swap transaction, by way of giving the dealer the right to cancel the swap.

[32] For a review of binary options, see Appendix A.

Issuer:	U.S. Agency	
Principal:	\$100mm	
Maturity:	15 years	
Coupon:	$X \times N/M$	
	where:	$X = 8\%$ for years 1–5
		$X = 12\%$ for years 6–10
		$X = 25\%$ for years 11–15
	and	$N =$ number of days in accrual period that Curve ≥ 0
		$M =$ number of days in accrual period
	Curve = 30-year CMS − 10-year CMS	
Payments:	Quarterly	
Day Count:	30/360	
Resets:	Daily for 30-year CMS and 10-year CMS for purposes of calculating the number of days the coupon is accrued	
Issue Price:	100%	
Call Feature:	This bond is callable quarterly starting in three months	

Table 7.14: Terms of a Step-Up Non-Inversion Note

And the issuer will normally take its cue to call the bond based on the dealer's decision to cancel the swap.

The dealer will cancel the swap depicted in Figure 7.15 when it is advantageous for him to do so. Assuming the curve is steep (in which case the forward differential between 30-year CMS and 10-year CMS is usually steep as well), the dealer will typically cancel the swap in a low rate environment and not cancel in a high rate environment. If the curve is steep, then there is a high probability that the dealer is paying out a coupon of 8.50% versus receiving 3-month LIBOR. The dealer may have economic incentive to cancel this trade when rates are low. If instead the curve is flat, then the forward differential between 30-year CMS and 10-year CMS will often be flat, and most pricing models will put a lot of weight on the possibility of the dealer paying out a coupon of 0%. In this case a pricing model will likely indicate that the dealer should not cancel the swap.

The callability of non-inversion notes is a powerful feature that allows for bonds with high (potential) coupons to be structured. For example, consider a bond whose terms are in Table 7.14. Lest we think this bond is fictitious, a bond similar to this one was issued in 2004. At the time, the 10-year swap rate was around 4.50%, and the 30-year swap rate was around 5.25% (and